

Exact Physically Assisted Quadrature of the Circle by the Hydrogeometric Transfer Method

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April 15, 2026

Abstract

We introduce the Hydrogeometric Transfer Method, an exact physically assisted procedure for the quadrature of the circle. The construction uses two vessels together with one additional operation: the exact transfer of water from one vessel to the other. It is important to note all measurements needed to construct the vessels can be carried out with straightedge and compass alone. In this way, the transcendental length $l = r\sqrt{\pi}$ is realised as a measurable water level. The method extends geometric construction by incorporating volumetric transfer as an exact operational step. Its chief merit is its remarkable simplicity: the passage from circle to square is governed by a single conservation law.

1 Introduction

The problem of squaring the circle is one of the classical problems of geometry. Given a circle of radius r , one seeks a square of equal area. In the strict Euclidean setting, where only compass and straightedge constructions are allowed, the problem is impossible: Lindemann proved in 1882 that π is transcendental, and consequently the length $r\sqrt{\pi}$ cannot be obtained by finitely many Euclidean construction steps.

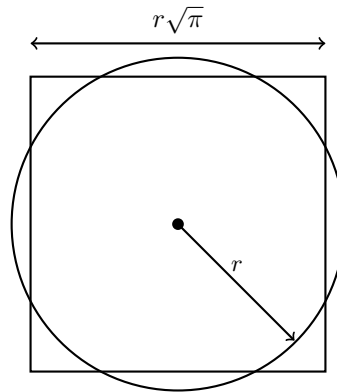


Figure 1: A circle of radius r and a square of side $r\sqrt{\pi}$ have the same area πr^2 .

This note presents what we call the Hydrogeometric Transfer Method: a physically assisted geometric procedure based on two vessels and the conservation of volume. The method renders the relation $l = r\sqrt{\pi}$ between a circle of radius r and an equal-area square of side l both visible and operational by means of an exact transfer of water between two constructible vessels.

2 The Hydrogeometric Transfer Method

The method rests on a single simple idea: conservation of volume converts the area of a circle into a measurable length inside a triangle.

The Two Aquariums

Aquarium 1 — Cylindrical aquarium. The cylinder is oriented so that its circular cross-section faces the viewer directly: the front face is a circle of radius r and the back face is an identical circle at depth h . The axis of

the cylinder is therefore horizontal, pointing away from the viewer. Let r be the radius of that circle and h the depth of the aquarium. The area of the circular base is

$$A = \pi r^2,$$

and the volume of the full cylinder is

$$V = A \cdot h.$$

Aquarium 2 — Triangular prism. The prism is oriented so that its triangular cross-section faces the viewer directly. The base is a right isosceles triangle with the hypotenuse at the top and in a strictly horizontal position; the right-angle vertex points downward. Each leg of the triangle equals the diameter $2r$ of the circle, and the depth of the prism equals the same h as the cylindrical aquarium. The area of the triangular base is

$$A_{\triangle} = \frac{1}{2}(2r)^2 = 2r^2,$$

and the volume of the full prism is

$$V_{\triangle} = A_{\triangle} \cdot h = 2r^2 h.$$

All dimensions required to build these two vessels and place them as described can be constructed exactly using only a straightedge and compass.

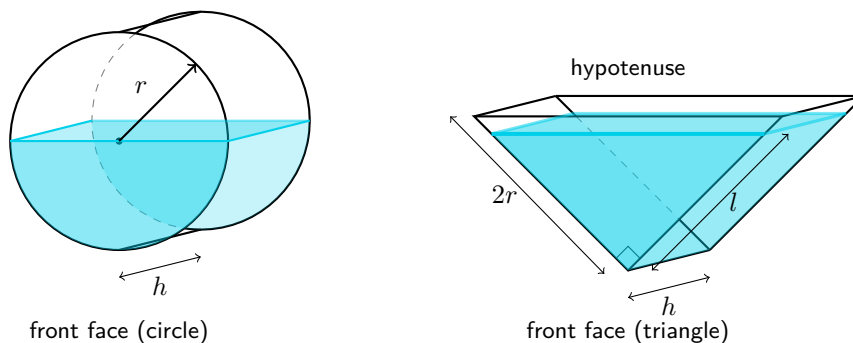


Figure 2: Left: cylindrical aquarium (front view is a circle of radius r , depth h). Right: triangular-prism aquarium (front view is a right isosceles triangle with hypotenuse on top, legs $= 2r$, depth h).

The Transfer Argument

Fill the cylindrical aquarium to exactly half its capacity. The water occupies a half-cylinder whose volume is

$$V_{\text{water}} = \frac{1}{2} A_{\text{circle}} \cdot h = \frac{1}{2} \pi r^2 h.$$

Now pour all of this water into the triangular prism. Because both containers share the same height h , the water level in the prism rises until the cross-sectional area it occupies equals

$$A_{\text{water in prism}} = \frac{V_{\text{water}}}{h} = \frac{1}{2} \pi r^2.$$

The key observation is:

$$A_{\text{water in prism}} = \frac{1}{2} A_{\text{circle}}.$$

Connecting Triangle to Circle

The procedure above shows that the area of the right isosceles triangle whose legs have a specific length l satisfies

$$\frac{1}{2} l^2 = \frac{1}{2} \pi r^2 \implies l = r\sqrt{\pi}.$$

In other words, if we construct the right isosceles triangle with legs $l = r\sqrt{\pi}$, its area equals $\frac{1}{2} \pi r^2$.

Reflecting that triangle across its hypotenuse, as in Figure 3, gives a square with side $l = r\sqrt{\pi}$ and area

$$l^2 = \pi r^2 = A_{\text{circle}},$$

which is exactly the area of the original circle.

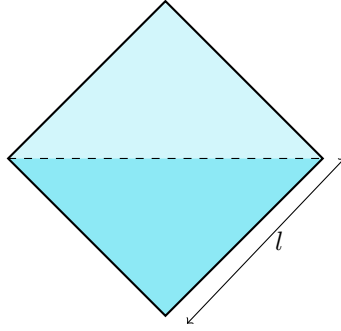


Figure 3: The water triangle and its reflection across the hypotenuse form a square of side l .

3 Conclusions

We set out to solve the quadrature problem for a given circle of radius r while allowing one auxiliary physical operation in addition to Euclidean construction. The Hydrogeometric Transfer Method proceeds in three steps:

1. Build two aquariums of the same depth h : a cylinder with circular cross-section of radius r , and a right-isosceles-triangle prism with legs equal to the diameter $2r$. Both are constructible with straightedge and compass.
2. Fill the cylinder to exactly half its capacity and pour all the water into the triangular prism.
3. The water level in the prism marks a height l along each leg of the triangle, where $l = r\sqrt{\pi}$.

The right isosceles triangle formed by the water surface has legs l and area $\frac{1}{2}l^2 = \frac{1}{2}\pi r^2$. Two such triangles joined along the hypotenuse give a square of side $l = r\sqrt{\pi}$ and area πr^2 — exactly the area of the original circle. The quadrature problem is therefore solved exactly within this physically assisted framework.

Its conceptual appeal lies precisely in that simplicity. No iterative approximation is required, no elaborate algebraic machinery is concealed in the argument, and no complicated measuring protocol is introduced. A single physical transfer, governed by conservation of volume, yields the length required for quadrature.